

Notes: Pinter, Wang & Zou (RFS, 2024)

Size Discount and Size Penalty: Trading Costs in Bond Markets

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1 Big picture

- **Question.** How should we understand the empirical relation between trade size and trading costs in over-the-counter bond markets?
- **Main tension.** The literature often documents a *size discount*: larger trades appear to receive lower trading costs. The paper shows that this pooled pattern can hide a *size penalty*: for the same client, larger trades can cost more.
- **Core distinction.** Trade size is not the same as trader size. Large clients tend to trade larger amounts and obtain better prices, generating the pooled size discount. But within a given client, unusually large trades are more expensive, generating the size penalty.
- **Data advantage.** The ZEN transaction data from the U.K. Financial Conduct Authority identify both sides of client-dealer trades, transaction time, price, quantity, ISIN, account number, and buy-sell flag.
- **Main sample.** The sample covers August 2011 to December 2017 and contains about 1.2 million U.K. government bond client-dealer trades, plus a similarly large corporate bond sample used for cross-market tests.
- **Main result.** In pooled government bond regressions, one log-point larger trades have lower costs: the coefficient is -0.217 bps. After client fixed effects, the sign flips. In the most restrictive government bond specification, one log-point larger trades cost 0.158 bps more.
- **Economic size.** A one-standard-deviation increase in government bond trade size, about 2.72 log points, raises trading costs by roughly 0.43 bps in the baseline within-client specification.
- **Mechanism.** The size penalty is larger for more sophisticated clients, stronger when trade size contains predictive information about returns or order flow, and much stronger in corporate bonds.
- **Identification idea.** The paper does not claim to estimate a clean causal effect of quantity on price. Instead, it uses client identity, client-month fixed effects, dealer-day fixed effects, bond-day fixed effects, client-dealer fixed effects, client-type heterogeneity, macro-news periods, relationship strength, and cross-market variation to separate bargaining/search forces from information forces.
- **Economic takeaway.** The same market can display a size discount across traders and a size penalty within traders. The pooled cost-size relation is therefore not a structural law of liquidity; it is the outcome of heterogeneous trader size, bargaining power, relationship strength, and information.

2 Data and measurement

2.1 ZEN transaction data

- The empirical setting is the U.K. bond market.
- The ZEN database records transaction-level information, including counterparty identities on both sides of the trade.
- The government bond analysis focuses on client trades with Gilt-Edged Market Makers, the primary dealers in the U.K. gilt market.
- After filters for duplicates and erroneous observations, the government bond sample contains roughly 1.2 million client-dealer trades.
- The authors also use a corporate bond sample of similar size to compare government and corporate bond markets.

Interpretation: The counterparty identifiers are the key data feature. Without client identity, the pooled cost-size relation confounds large trades with large traders.

2.2 Client sophistication

The paper classifies clients into two broad groups:

- **More sophisticated clients:** hedge funds and asset managers.
- **Less sophisticated clients:** pension funds, foreign central banks, commercial banks, international policy institutions, insurance companies, and nonfinancial investors.

Interpretation: The classification is designed to separate clients more likely to trade on information from clients more likely to trade for hedging, duration, or portfolio-allocation motives.

2.3 Trade size

The trade-size variable is the natural logarithm of the pound notional value of a transaction:

$$Size_v = \log(\text{notional value of trade } v \text{ in pounds}). \quad (1)$$

Government bond trades are larger than corporate bond trades. In the histograms, government bonds have mean log size 13.4 and median 13.7, while corporate bonds have mean 12.1 and median 12.2.

Interpretation: Using logs lets the coefficient be interpreted as the bps cost change associated with proportional changes in trade size.

2.4 Trading cost measure

For each trade v , the paper follows O'Hara and Zhou-style signed price deviations:

$$Cost_v = [\ln(P_v^*) - \ln(\bar{P})] \times \mathbf{1}_{B,S}. \quad (1)$$

Here P_v^* is the transaction price, $\bar{P}$ is the benchmark price, and $\mathbf{1}_{B,S}$ equals +1 for buys and -1 for sells. The paper multiplies the measure by 10,000, so the reported cost is in basis points.

Interpretation: A buyer paying above the benchmark has a positive cost. A seller receiving below the benchmark also has a positive cost because the sell indicator reverses the sign.

2.5 Summary-statistic logic

- In pooled medians, larger government bond trades often look cheaper than small trades.
- After removing client-specific averages, the pattern reverses: larger trades often have higher costs.
- This is the first visual indication that cross-client and within-client variation are pointing in different directions.

Interpretation: The size discount comes from comparing different clients. The size penalty comes from comparing large and small trades by the same client.

3 Models and empirical specifications

3.1 Baseline cost-size regression

The core trade-level regression is

$$Cost_v = \beta Size_v + \alpha_{k,t} + \lambda_{i,m} + \mu_{j,t} + \delta_{i,j} + \varepsilon_v. \quad (2)$$

The fixed effects can include:

- bond-day fixed effects $\alpha_{k,t}$;
- client-month fixed effects $\lambda_{i,m}$;
- dealer-day fixed effects $\mu_{j,t}$;
- client-dealer fixed effects $\delta_{i,j}$.

Interpretation: The coefficient β asks whether larger trades have higher or lower costs after comparing trades within a narrow fixed-effect cell. Without client fixed effects, β captures cross-client and within-client variation together. With client fixed effects, β moves toward the within-client size penalty.

3.2 Cross-sectional trader-size regression

To isolate the cross-client size discount, the authors collapse the data to the client level and estimate

$$Cost_i = c + \gamma Size_i + \varepsilon_i, \quad (3)$$

where $Cost_i$ is average client trading cost and $Size_i$ is the logarithm of the client's average trade size.

Interpretation: A negative γ means larger clients obtain lower average trading costs. This is the trader-size channel behind the pooled size discount.

3.3 Client-type heterogeneity

To compare more and less sophisticated clients, the paper estimates

$$Cost_v = \sum_{w=1}^2 \eta_w \mathbf{1}\{i \in g_w\} Size_v + FE + \varepsilon_v. \quad (4)$$

The groups are more sophisticated and less sophisticated clients.

Interpretation: If the size penalty is driven by information, it should be larger for sophisticated clients. If it is driven only by dealer inventory costs, the client-type difference should be weaker.

3.4 Macro-news heterogeneity

The paper interacts trade size with client type and a dummy for large macroeconomic surprises. Conceptually,

$$Cost_v = \eta_{w,n} \mathbf{1}\{i \in g_w, t \in n\} Size_v + FE + \varepsilon_v, \quad (2)$$

where n indexes small-news and large-news trading days.

Interpretation: Information asymmetry should be more important on information-intensive days. Therefore, a larger size penalty around macro news supports an information-based interpretation.

3.5 Return-predictability test

For client type k and horizon T , the paper defines

$$ExcessReturn_{k,t}^T = LargeSizeReturn_{k,t}^T - SmallSizeReturn_{k,t}^T. \quad (6)$$

The large-size and small-size portfolios mimic order flow from clients of type k when those clients trade in unusually large versus small sizes.

Interpretation: If large trades by sophisticated clients contain information, the large-size portfolio should outperform the small-size portfolio.

3.6 Order-flow predictability

For bond j and day t , the paper estimates

$$OrderFlow_{t+1,t+d}^U = \alpha_t + \delta_j + \beta OrderFlow_{t-4,t}^S + \varepsilon_{j,t}. \quad (7)$$

Then it splits sophisticated-client order flow into small- and large-size components:

$$OrderFlow_{t+1,t+d}^U = \alpha_t + \delta_j + \beta_1 SmallSizeOrderFlow_{t-4,t}^S + \beta_2 LargeSizeOrderFlow_{t-4,t}^S + \varepsilon_{j,t}. \quad (8)$$

Interpretation: The information interpretation predicts $\beta_2 > \beta_1$: large-size order flow of sophisticated clients should forecast future order flow of less sophisticated clients more strongly.

3.7 Relationship strength

Client-side relationship strength is measured as the share of client i 's previous-six-month trading volume with dealer m :

$$\Omega_{i,m,t-6 \rightarrow t}^c = \frac{\sum_{j_1=1}^{N_{i,m,t-6 \rightarrow t}} quantity_{j_1}}{\sum_{j_2=1}^{N_{i,t-6 \rightarrow t}} quantity_{j_2}}. \quad (10)$$

Dealer-side relationship strength is defined analogously:

$$\Omega_{i,m,t-6 \rightarrow t}^g = \frac{\sum_{j_1=1}^{N_{i,m,t-6 \rightarrow t}} quantity_{j_1}}{\sum_{j_2=1}^{N_{m,t-6 \rightarrow t}} quantity_{j_2}}. \quad (11)$$

Interpretation: Stronger relationships can reduce asymmetric information and therefore reduce the size penalty.

3.8 Government versus corporate bond markets

For market z and client group w , the paper estimates

$$Cost_v = \sum_{w=1}^2 \sum_{z=1}^2 \eta_{w,z} \mathbf{1}\{j \in l_z, i \in g_w\} Size_v + FE + \varepsilon_v. \quad (12)$$

Interpretation: Corporate bonds are less liquid and more opaque than government bonds. A larger size penalty in corporate bonds, especially for sophisticated clients, is consistent with an information channel.

4 Identification and empirical design

4.1 What the paper identifies

- The paper separates two empirical objects: a pooled cost-size relation and a within-client cost-size relation.
- The pooled relation answers: do larger transactions, which are often done by larger clients, receive lower costs?
- The within-client relation answers: when the same client trades unusually large, is the trade more expensive?
- The paper is explicit that it does not estimate a randomized causal effect of trade size on transaction cost.

Interpretation: The goal is to decompose sources of variation and mechanisms, not to claim that trade size is randomly assigned.

4.2 Controls for alternative mechanisms

- **Client-month fixed effects** absorb low-frequency changes in client size and average informativeness.
- **Dealer-day fixed effects** absorb time-varying dealer balance sheet constraints and inventory conditions.
- **Bond-day fixed effects** absorb market conditions specific to a bond on a given day.
- **Client-dealer fixed effects** absorb persistent relationship quality and bargaining arrangements.
- **Client-type interactions** compare sophisticated and less sophisticated clients.
- **Macro-news splits** use periods when information asymmetry should be more important.
- **Corporate-government comparisons** use differences in opacity and liquidity across markets.

Interpretation: Each layer narrows the interpretation: the size penalty is not just a pooled bargaining discount, not just bond-level liquidity, not just dealer-day balance-sheet pressure, and not just persistent relationship quality.

4.3 Why client identity is central

- Without client identifiers, one cannot tell whether large trades are cheaper because they are large or because they are executed by large clients.
- Client fixed effects remove each client's average cost and trade-size level.
- This converts the question from cross-sectional bargaining to within-client trade-size variation.

Interpretation: Client identity is the empirical device that resolves the apparent contradiction between size discount and size penalty.

4.4 What remains imperfect

- Trade size and trading costs are jointly determined in equilibrium.
- Private information, liquidity shocks, and inventory pressure are not directly observed.
- The classification of sophisticated versus less sophisticated clients is useful but coarse.
- The paper relies on a set of fixed effects and heterogeneity tests rather than a single clean instrument.

Interpretation: The evidence is strongest as a mechanism-based decomposition rather than as a standalone causal estimate.

5 Tables and figures

This section keeps the same *Read* plus *Economic message* structure. Adjacent figures and tables are grouped to reduce page breaks and speed up comparison.

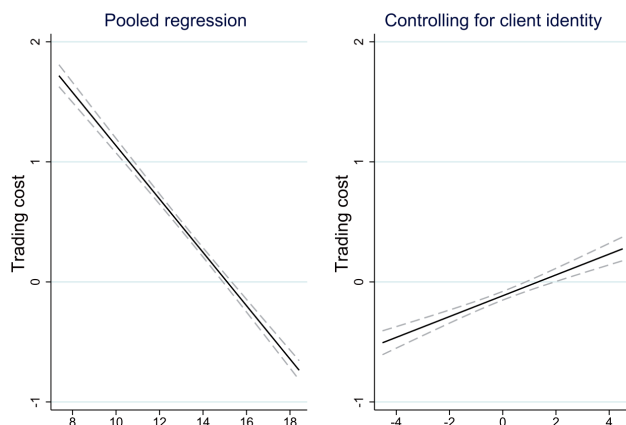
5.1 Figure 1 + Figure 2: the basic puzzle and trade-size distributions

Read:

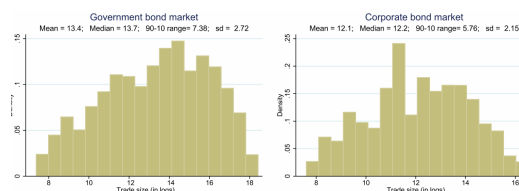
- Figure 1 shows two different cost-size slopes in government bonds.
- In the pooled regression, larger trades have lower trading costs: the left panel slopes down.
- After controlling for client identity, the slope turns positive: the right panel shows a within-client size penalty.
- Figure 2 shows that government bond trades are larger and more dispersed than corporate bond trades.

- Government bond log size has mean 13.4 and standard deviation 2.72; corporate bond log size has mean 12.1 and standard deviation 2.15.

Economic message: The first figure is the paper in one picture. The same data show a size discount across clients and a size penalty within clients.



(a) Figure 1: pooled size discount versus within-client size penalty.



(b) Figure 2: trade-size distributions.

5.2 Table 1 + Table 2: summary medians and baseline regressions

Read:

- Table 1 reports median trading costs by trade-size and maturity bins.
- In pooled medians, many larger trade-size bins have lower costs than smaller bins.
- After removing client-specific averages, larger trades tend to have higher costs.
- Table 2 formalizes this pattern in regressions.
- Column 1 gives a pooled size discount: -0.217 bps per log-point of trade size.
- Column 2, after client fixed effects, flips the sign to $+0.102$ bps.
- Column 5, with day-bond, month-client, day-dealer, and client-dealer controls, estimates $+0.158$ bps.

Economic message: The sign flip is the central empirical result. Large traders get better prices, but the same trader pays more when trading larger amounts.

Table 1
Summary statistics: Trading costs, trade size, and maturity

(a) Results from the pooled data set

Maturity	Trading cost			Number of obs.		
	0-8Y	8-20Y	>20Y	0-8Y	8-20Y	>20Y
Trade-size category						
£1–£100,000	0.778	1.447	0.530	157,533	87,631	89,004
£100,000–£1,000,000	–0.117	–0.097	–0.289	123,848	89,900	107,341
£1,000,000–£5,000,000	–0.267	–0.374	–0.420	102,478	91,682	81,865
>£5,000,000	–0.021	–0.171	0.200	145,555	108,783	88,675

(b) After removing client-specific averages

Maturity	Trading cost			Number of obs.		
	0-8Y	8-20Y	>20Y	0-8Y	8-20Y	>20Y
Trade-size category						
£1–£100,000	–0.627	–0.115	–0.072	157,533	87,631	89,004
£100,000–£1,000,000	–0.528	–0.393	–0.395	123,848	89,900	107,341
£1,000,000–£5,000,000	–0.240	–0.051	–0.173	102,478	91,682	81,865
>£5,000,000	0.146	0.048	0.381	145,555	108,783	88,675

Original Table 1: summary statistics for trading costs, trade size, and maturity.

Table 2
Trading costs and trade size in government bond markets: The role of traders' identity

	(1)	(2)	(3)	(4)	(5)
Trade size	-0.217*** (-4.04)	0.102*** (3.05)	0.121*** (4.01)	0.134*** (4.25)	0.158*** (5.19)
N	1,274,295	1,274,289	1,274,289	1,269,855	1,269,238
R ²	.055	.061	.062	.134	.139
Day#Bond FE	Yes	Yes	Yes	Yes	Yes
Client FE	No	Yes	Yes	No	No
Dealer FE	No	No	Yes	No	No
Day#Dealer FE	No	No	No	Yes	Yes
Month#Client FE	No	No	No	Yes	Yes
Client#Dealer FE	No	No	No	No	Yes

Original Table 2: baseline cost-size regressions in government bonds.

5.3 Figure 3 + Table 3: cross-sectional size discount and client sophistication

Read:

- Figure 3 collapses the data to the client level and shows that larger clients have lower average trading costs.
- The estimated cross-sectional slope is about -0.59 with a large negative t -statistic.
- Table 3 separates less sophisticated and more sophisticated clients.
- The size penalty in the most restrictive specification is 0.106 bps for less sophisticated clients and 0.197 bps for more sophisticated clients.
- The point estimates imply a larger size penalty for sophisticated clients, although the equality test is not always significant in the most restrictive columns.

Economic message: The cross-section delivers the size discount. Client-type heterogeneity points toward information as a source of the within-client size penalty.

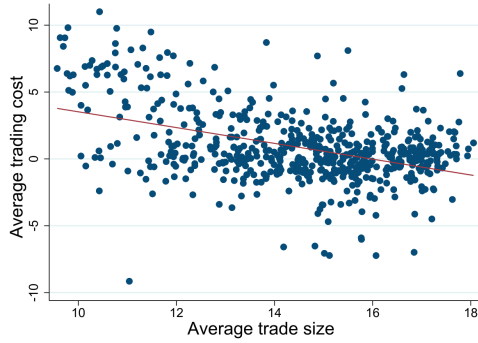


Figure 3

(a) Figure 3: average cost versus average client trade size.

Table 3
Trading costs and trade size: More sophisticated clients versus less sophisticated clients

	(1)	(2)	(3)	(4)	(5)
Less sophisticated clients					
Trade size	-0.217*** (-4.14)	0.057 (1.22)	0.067 (1.46)	0.080* (1.79)	0.106** (2.29)
More sophisticated clients					
Trade size	-0.213*** (-2.67)	0.140*** (3.34)	0.169*** (4.83)	0.183*** (5.08)	0.197*** (5.61)
p-values, eq. of coeff.	.966	.185	.076	.069	.115
N	1,271,112	1,271,106	1,271,106	1,264,580	1,263,963
R ²	.100	.106	.107	.202	.207
Day#Bond#ClientType FE	Yes	Yes	Yes	Yes	Yes
Client FE	No	Yes	Yes	No	No
Dealer#ClientType FE	No	No	Yes	No	No
Day#Dealer#ClientType FE	No	No	No	Yes	Yes
Month#Client FE	No	No	No	Yes	Yes
Client#Dealer FE	No	No	No	No	Yes

(b) Table 3: more versus less sophisticated clients.

5.4 Table 4 + Figures 4–5: macro news and return predictability

Read:

- Table 4 interacts trade size with macro-news periods and client type.
- For more sophisticated clients, the size penalty is higher on large-news days than on small-news days in every specification shown.
- Figure 4 constructs long-short portfolios that mimic large-size versus small-size order flow.
- The strategy is not significant for less sophisticated clients and is often negative.

- For more sophisticated clients, the strategy earns positive cumulative returns that reach roughly 8 bps at the 30-day horizon.
- Figure 5 decomposes sophisticated-client returns by macro-news size. The positive return predictability is stronger around large macroeconomic surprises.

Economic message: Large trades by sophisticated clients appear to carry information. The size penalty is connected not only to execution frictions but also to future return predictability.

Table 4
Trading costs and trade size: Around big and small macroeconomic news

	(1)	(2)	(3)	(4)	(5)
Less sophisticated clients					
Trade size#SmallNews	0.061 (1.25)	0.080 (1.57)	0.061 (1.33)	0.082* (1.70)	0.104** (2.11)
Trade size#LargeNews	0.067 (1.44)	0.069 (1.40)	0.066 (1.41)	0.075 (1.58)	0.107** (2.13)
p-values, eq. of coeff.	.830	.673	.871	.833	.943
More sophisticated clients					
Trade size#SmallNews	0.150*** (4.01)	0.161*** (4.03)	0.141*** (3.49)	0.154*** (3.62)	0.171*** (4.11)
Trade size#LargeNews	0.196*** (4.55)	0.207*** (4.75)	0.202*** (4.77)	0.205*** (4.79)	0.221*** (5.17)
p-values, eq. of coeff.	.132	.119	.086	.144	.147
N	1,182,307	1,178,836	1,179,827	1,176,302	1,175,687
R ²	.106	.136	.173	.199	.204
Day#Bond#ClientType FE	Yes	Yes	Yes	Yes	Yes
Dealer#ClientType FE	Yes	No	No	No	No
Client FE	Yes	Yes	Yes	No	No
Day#Dealer#ClientType FE	No	No	Yes	Yes	Yes
Month#Client FE	No	Yes	No	Yes	Yes
Client#Dealer FE	No	No	No	No	Yes

Original Table 4: size penalty around big and small macroeconomic news.

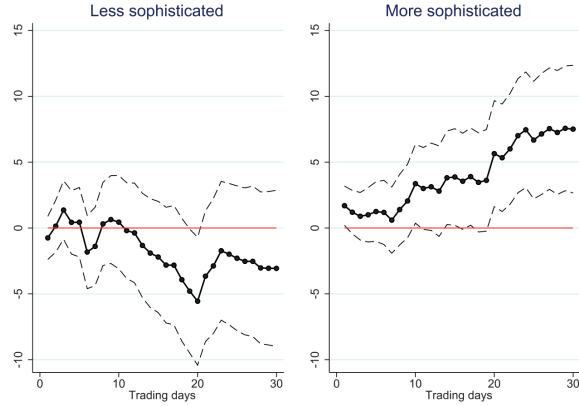


Figure 4

(a) Figure 4: large-size minus small-size return portfolios by client type.

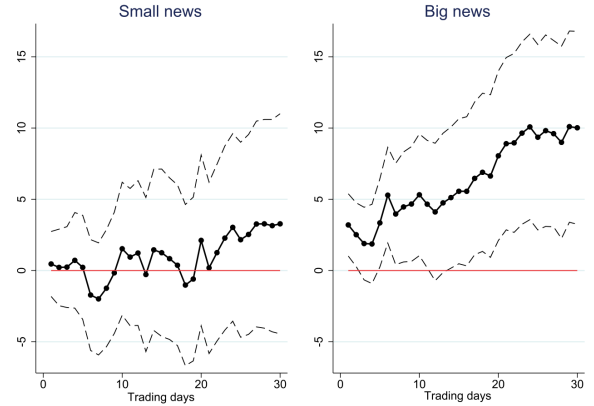


Figure 5

(b) Figure 5: large versus small macro-news periods.

5.5 Table 5 + Table 6: future order flow and relationship concentration

Read:

- Table 5 shows that sophisticated clients' recent order flow predicts less sophisticated clients' future order flow.
- In panel A, $OrderFlow_{t-4,t}^S$ significantly predicts next-day and next-week order flow.
- In panel B, the predictability is concentrated in large-size order flow: small-size order flow is generally insignificant, while large-size order flow is positive and significant.

- Table 6 reports relationship concentration.
- About 29.55% of trades occur with a client's most-favored dealer, and 44.54% occur with one of a client's two most-favored dealers.
- From the dealer side, only 8.60% of trades are with the dealer's most-favored client and 14.98% with one of its two most-favored clients.

Economic message: Large trades by sophisticated clients forecast future flow, and relationship strength matters because trading is not anonymous. OTC trading costs depend on who trades with whom.

Table 5
Predicting future order flow

	(1)	(2)	(3)	(4)	(5)	(6)
	<i>Order Flow</i> ^U _{t+1,t+1}			<i>Order Flow</i> ^U _{t+1,t+5}		
(a) The predictive power of order flow						
<i>Order Flow</i> ^S _{t-4,t}	0.026*** (4.17)	0.026*** (4.02)	0.023*** (3.73)	0.069*** (3.21)	0.071*** (3.33)	0.055*** (2.93)
N	62,249	62,249	62,249	58,761	58,761	58,761
R ²	.001	.025	.038	.001	.024	.063
Day FE	No	Yes	Yes	No	Yes	Yes
Instrument FE	No	No	Yes	No	No	Yes
(b) Small-sized versus large-sized order flow						
<i>Small Size OF</i> ^S _{t-4,t}	0.005 (0.40)	0.004 (0.34)	0.007 (0.61)	-0.031 (-0.61)	-0.039 (-0.76)	-0.025 (-0.47)
<i>Large Size OF</i> ^S _{t-4,t}	0.010** (2.40)	0.010** (2.54)	0.007* (1.91)	0.050*** (3.40)	0.053*** (3.66)	0.037*** (2.83)
N	57,336	57,336	57,336	54,636	54,636	54,636
R ²	.000	.027	.039	.001	.028	.064
Day FE	No	Yes	Yes	No	Yes	Yes
Instrument FE	No	No	Yes	No	No	Yes

(a) Table 5: predicting future order flow.

Table 6
Distribution of trades across top client-dealer relationships

	(1) Yes	(2) No
Clients' most-favored dealer	29.55%	70.45%
Clients' two most-favored dealers	44.54%	55.46%
Dealers' most-favored client	8.60%	91.40%
Dealers' two most-favored clients	14.98%	85.02%

Column 1 of this table presents the ratio of number of transactions that are executed via favourite client-dealer relationships to the total number of transactions in our sample (with column 2 presenting the fraction of trades outside these relationships). We use four definitions of favourite relationships based on measures 10 and 11.

(b) Table 6: distribution of top client-dealer relationships.

5.6 Table 7 + Table 8: relationship strength and government versus corporate bonds

Read:

- Table 7 shows that the size penalty is larger in weaker relationships.
- For clients' weaker dealer relationships, the column 4 coefficient is 0.202 bps; for the strongest dealer relationship it is 0.073 bps.
- For dealers' weaker client relationships, the column 4 coefficient is 0.173 bps; for strongest client relationships it is 0.054 bps.
- Table 8 compares government and corporate bonds.
- For less sophisticated clients, the corporate-government difference is not statistically strong in the most restrictive specification.
- For more sophisticated clients, the corporate bond size penalty is 0.856 bps versus 0.186 bps for government bonds, with a near-zero equality-test p -value.

Economic message: Strong relationships mitigate the size penalty, and the penalty is much larger in more opaque corporate bond markets for sophisticated clients. Both patterns reinforce the information-based interpretation.

Table 7
Trading costs and trade size: The role of relationship strength

	(1)	(2)	(3)	(4)
<i>A. Clients' weak versus strong (top-1) dealer relationship</i>				
<i>Clients' weaker dealer relationships</i>				
Trade size	0.138*** (4.05)	0.153*** (4.84)	0.177*** (5.97)	0.202*** (7.06)
<i>Client's strongest (top-1) dealer relationships</i>				
Trade size	0.050 (1.09)	0.088** (2.42)	0.071* (1.83)	0.073* (1.80)
p-values, eq. of coeff.	.036	.031	.003	.001
N	1,177,267	1,177,267	1,169,789	1,169,163
R ²	.107	.107	.198	.203
<i>B. Dealers' weak versus strong (top-1) client relationship</i>				
<i>Dealers' weaker client relationships</i>				
Trade size	0.116*** (3.28)	0.140*** (4.55)	0.150*** (4.76)	0.173*** (5.71)
<i>Dealers' strongest (top-1) client relationships</i>				
Trade size	−0.026 (−0.39)	0.009 (0.14)	0.054 (0.49)	0.054 (0.49)
p-values, eq. of coeff.	.048	.049	.415	.310
N	1,166,020	1,166,020	1,156,271	1,155,658
R ²	.084	.084	.165	.170
Day#Bond#ClientType FE	Yes	Yes	Yes	Yes
Client FE	Yes	Yes	No	No
Dealer#ClientType FE	No	Yes	No	No
Day#Dealer#ClientType FE	No	No	Yes	Yes
Month#Client FE	No	No	Yes	Yes
Client#Dealer FE	No	No	No	Yes

Original Table 7: the role of relationship strength.

Table 8
Trading costs and trade size: Government versus corporate bonds markets

	(1)	(2)	(3)	(4)
<i>Less sophisticated clients</i>				
Trade size#GovernmentBonds	0.078 (1.42)	0.093** (2.01)	0.122*** (2.69)	0.125*** (2.74)
Trade Size#CorporateBonds	0.310 (1.59)	0.316 (1.59)	0.332 (1.64)	0.350* (1.71)
p-values, eq. of coeff.	.149	.192	.232	.208
<i>More sophisticated clients</i>				
Trade Size#GovernmentBonds	0.152*** (3.92)	0.177*** (4.70)	0.186*** (5.16)	0.186*** (5.11)
Trade Size#CorporateBonds	0.774*** (3.89)	0.749*** (3.81)	0.800*** (4.07)	0.856*** (4.34)
p-values, eq. of coeff.	0.001	0.002	0.001	0.000
N	1,171,526	1,165,674	1,165,359	1,164,790
R ²	.349	.426	.430	.433
Day#Bond#ClientType FE	Yes	Yes	Yes	Yes
Client#Market FE	Yes	Yes	Yes	No
Dealer#Market#ClientType FE	Yes	Yes	Yes	No
Day#Dealer#ClientType FE	No	Yes	Yes	Yes
Month#Client FE	No	Yes	Yes	Yes
Client#Dealer	No	No	Yes	No
Client#Dealer#Market FE	No	No	No	Yes

This table regresses trading costs (measured in bp) on trade size (measured as the logarithm of the nominal size of the trade in £s) interacted with an indicator variable taking value 2 (1) if the trade takes place in the corporate (government) market.

Original Table 8: government versus corporate bond markets.

6 Short summary

- The paper studies the relation between trade size and trading costs in OTC bond markets.
- The empirical puzzle is that larger trades often appear cheaper in pooled data, but theories of asymmetric information and inventory risk can predict higher costs for larger trades.
- The paper resolves the puzzle by distinguishing trader size from trade size.
- Large clients trade large amounts and receive better prices, generating the size discount across clients.

- The same client pays more when executing unusually large trades, generating the size penalty.
- The ZEN database is crucial because it identifies clients and dealers on both sides of a trade.
- The baseline government bond regression flips from a negative coefficient to a positive coefficient once client identity is controlled for.
- The size penalty is larger for more sophisticated clients, especially in information-sensitive settings.
- Large trades by sophisticated clients predict future returns and future order flow.
- Strong client-dealer relationships weaken the size penalty.
- Corporate bond markets display a much larger size penalty than government bond markets for sophisticated clients.
- The paper’s evidence supports both imperfect-competition theories and information-based theories, but each applies to a different source of variation.

7 Conclusion

7.1 Core conclusion

Pinter, Wang, and Zou show that the apparent size discount in bond markets is mainly a cross-client phenomenon. Large clients tend to receive better prices and make larger trades. After holding client identity fixed, larger trades are more expensive. This within-client size penalty is strongest where information asymmetry is more plausible: sophisticated clients, large macro-news periods, weaker relationships, and corporate bond markets.

7.2 Economic intuition

The key intuition is that OTC prices reflect both bargaining and information. Larger clients have more bargaining power or search intensity, so they receive better average prices. But when a given client trades unusually large, the trade may reveal information or impose adverse-selection risk on dealers. Dealers respond by charging higher costs. A pooled regression mixes these forces and can produce the wrong economic interpretation.

7.3 Takeaway

The cost-size relation is not one object. Across traders, it reflects bargaining power and client size. Within traders, it reflects information, urgency, and relationship-specific frictions. Empirically, separating these margins requires client identity and rich fixed effects. The paper’s main lesson is that large trades are cheap only when they are large because the trader is large; when the same trader trades larger than usual, the trade becomes costly.